China Economic Sentiment Index: Construction Methodology and Technical Documentation

Economic Index Research Team

April 2026

Abstract

This document describes the construction methodology of the China Economic Sentiment Index (经济财经舆情指数), a monthly composite indicator measuring economic and financial sentiment in China. The index combines NLP-based news sentiment analysis, keyword frequency tracking across Chinese financial media, and principal component analysis (PCA) of seven sub-indicators drawn from text data, macroeconomic series, and internet search trends. The index is scaled to $[0, 100]$, with 50 representing neutral sentiment. We provide full transparency on data sources, text processing, sub-indicator construction, and the PCA aggregation procedure to ensure reproducibility.

1 Introduction

Timely measurement of economic sentiment is valuable for policymakers, investors, and researchers seeking to understand shifts in expectations before they appear in official statistics. For China, the world's second-largest economy, sentiment tracking is particularly important given the speed at which policy signals propagate through financial media and the limited frequency of many official data releases.

This document describes the China Economic Sentiment Index (CESI), a monthly composite index designed to provide a transparent, reproducible measure of China's economic and financial sentiment. The index ranges from 0 to 100, where 50 denotes neutral sentiment, values above 50 indicate net optimism, and values below 50 indicate net pessimism.

The index is updated monthly and combines three layers of information: (i) NLP-based sentiment scores derived from Chinese financial news headlines, (ii) keyword frequency analysis tracking the prevalence of positive, negative, and uncertainty-related economic terms, and (iii) macroeconomic and attention indicators sourced from FRED and Google Trends. These sub-indicators are aggregated via principal component analysis into a single composite score.

The CESI contributes to the growing literature on text-based economic indicators by focusing specifically on Chinese-language financial media. It complements existing measures such as the Purchasing Managers' Index (PMI), consumer confidence surveys, and the Baker–Bloom–Davis Economic Policy Uncertainty Index by offering a higher-frequency, media-derived perspective on sentiment.

The remainder of this document is organized as follows. Section 2 describes the data sources. Section 3 details the three-layer construction methodology. Section 4 presents backfilled results. Section 5 discusses storage and reproducibility. Section 6 provides an interpretation guide. Section 7 notes limitations, and Section 8 outlines the validation approach.

2 Data Sources

The index draws on three categories of data: news text from Chinese financial portals, macroeconomic indicators retrieved from the FRED API, and internet search attention data from Google Trends.

2.1 News Text Sources

We scrape headlines and lead paragraphs from two major Chinese financial news portals at the end of each calendar month.

Sina Finance (新浪财经). Headlines and lead paragraphs are collected from `finance.sina.com.cn` section pages covering macroeconomics, equities, and foreign exchange. Each monthly scrape yields approximately 100 articles.

East Money (东方财富). Headlines are scraped from `finance.eastmoney.com` listing pages spanning the macro, stock, and finance sections. Each monthly scrape yields approximately 100 articles.

Both sources are scraped once at month-end using Python `requests` and `BeautifulSoup`.

2.2 Macroeconomic Indicators

Four macroeconomic series are retrieved from the Federal Reserve Economic Data (FRED) API, as summarized in Table 1. Daily series are averaged to monthly frequency.

Table 1: Macroeconomic indicators from FRED

Series	FRED ID	Frequency	Description
USD/CNY Exchange Rate	`DEXCHUS`	Daily → Monthly avg	Spot exchange rate
CBOE VIX Index	`VIXCLS`	Daily → Monthly avg	Global risk appetite proxy
China CPI	`CHNCPIALLMINMEI`	Monthly	Consumer price index, all items
US Yield Curve	`T10Y2Y`	Daily → Monthly avg	10Y–2Y Treasury spread

2.3 Google Trends

We retrieve monthly Google Trends data for four queries: "China economy," "中国经济," "trade war China," and "China stimulus." Data are obtained at monthly resolution via the `pytrends` Python library. The attention sub-indicator is the arithmetic mean of the interest scores across all four queries.

3 Methodology

The index is constructed in three layers: NLP sentiment scoring (Section 3.1), keyword frequency analysis (Section 3.2), and PCA-based aggregation (Section 3.3).

3.1 Layer 1: NLP Sentiment Scoring

Each scraped article is processed through the following pipeline:

1. **HTML and URL removal.** Raw HTML tags and embedded URLs are stripped from the text.
2. **Chinese text segmentation.** The cleaned text is segmented into tokens using the `jieba` library.
3. **Stopword removal.** A standard Chinese stopword list is applied to remove function words and other uninformative tokens.
4. **Sentiment scoring.** Each article is scored using the `SnowNLP` library, which returns a score $s_j \in [0, 1]$, where 0 denotes very negative sentiment and 1 denotes very positive sentiment.

The monthly NLP sentiment sub-index is the arithmetic mean of all article-level scores in month t:

$$S_t = \frac{1}{N_t} \sum_{j=1}^{N_t} s_{j,t}, \tag{1}$$

where N_t is the number of articles scraped in month t.

3.2 Layer 2: Keyword Frequency Analysis

We track the occurrence of keywords in three categories within each month's corpus. All keywords are matched after `jieba` segmentation of the full text corpus.

Positive keywords (20). 经济复苏, 刺激政策, 消费回暖, 稳增长, 降准, 降息, 新质生产力, 高质量发展, 外资流入, 出口增长, 经济企稳, 回升向好, 逆周期调节, 扩大内需, 减税降费, 营商环境优化, 就业改善, 消费升级, 产业升级, 科技创新.

Negative keywords (20). 经济下行, 通缩, 失业, 债务危机, 资本外流, 房地产危机, 产能过剩, 制裁, 贸易战, 金融风险, 暴雷, 违约, 经济放缓, 通货膨胀, 滞胀, 资产泡沫, 系统性风险, 信用收缩, 外需疲软, 供应链中断.

Uncertainty keywords (12). 政策调整, 监管, 不确定性, 改革, 双循环, 共同富裕, 政策转向, 结构性改革, 宏观审慎, 去杠杆, 防风险, 政策博弈.

For each category, keyword occurrences are counted in the month's corpus and normalized by total word count (per 1,000 words). We define:

$$K_t^+ = \frac{\text{positive keyword count}_t}{\text{total word count}_t} \times 1{,}000, \tag{2}$$

$$K_t^- = \frac{\text{negative keyword count}_t}{\text{total word count}_t} \times 1{,}000, \tag{3}$$

$$K_t^{unc} = \frac{\text{uncertainty keyword count}_t}{\text{total word count}_t} \times 1{,}000. \tag{4}$$

The net keyword sentiment sub-indicator is:

$$K_t^{net} = K_t^+ - K_t^-. \tag{5}$$

3.3 Layer 3: PCA Composite Index

3.3.1 Input Variables

Seven sub-indicators enter the PCA:

1. S_t: NLP sentiment sub-index (Equation 1).
2. K_t^{net}: Net keyword frequency (Equation 5).
3. K_t^{unc}: Uncertainty keyword frequency (Equation 4).
4. ΔFX_t: USD/CNY monthly percentage change, inverted so that positive values indicate CNY appreciation.
5. VIX_t: Monthly average VIX level, inverted so that lower risk maps to higher values.
6. π_t^{YoY}: China CPI year-over-year percentage change.
7. GT_t: Google Trends attention index (Section 2.3).

3.3.2 Standardization

All sub-indicators are standardized using a rolling 24-month window:

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_{i,\,t-24:t}}{\sigma_{i,\,t-24:t}}, \tag{6}$$

where $\bar{x}_{i,\,t-24:t}$ and $\sigma_{i,\,t-24:t}$ denote the mean and standard deviation of sub-indicator i over the preceding 24 months (inclusive of month t).

For variables where higher values indicate worse sentiment—USD/CNY change showing depreciation, VIX, and keyword uncertainty—signs are inverted before standardization.

3.3.3 Principal Component Extraction

The composite index is derived from the first principal component (PC1) of the correlation matrix of the seven z-scored sub-indicators. Let $\mathbf{Z}_t = (z_{1,t}, \ldots, z_{7,t})'$. The correlation matrix is:

$$\mathbf{C} = \mathrm{Corr}(\mathbf{Z}). \tag{7}$$

We compute the eigendecomposition:

$$\mathbf{C} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}', \tag{8}$$

where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_7)$ with $\lambda_1 \geq \cdots \geq \lambda_7$, and $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_7]$.

The PC1 loading vector is $\mathbf{w} = \mathbf{v}_1$. The raw composite score for month t is:

$$PC1_t = \mathbf{z}_t' \mathbf{w}. \tag{9}$$

Sign convention. The sign of $\mathbf{w}$ is chosen such that $PC1_t$ is positively correlated with the NLP sentiment sub-index S_t.

3.3.4 Rescaling

The raw PC1 score is mapped to a $[0, 100]$ scale:

$$I_t = \max\left(0,\ \min\left(100,\ 50 + PC1_t \times \frac{50}{3}\right)\right). \tag{10}$$

This mapping assumes z-scores typically fall in $[-3, 3]$, centering the index at 50 (neutral).

3.3.5 Recalibration

PCA loadings are re-estimated quarterly using the trailing 24 months of data. Between recalibrations, new observations are projected onto the existing loading vector $\mathbf{w}$. All historical loadings are stored in the database for full reproducibility.

3.3.6 Interim Rule

Before 18 months of data are available—the minimum for stable PCA estimation—the composite is computed as:

$$I_t^{interim} = \max\left(0,\ \min\left(100,\ 50 + \bar{z}_t \times \frac{50}{3}\right)\right), \tag{11}$$

where $\bar{z}_t$ is the equal-weighted mean of all available z-scored sub-indicators.

4 Backfilled Results

We backfill the index from January 2023 through March 2025 using archived news data and historical macroeconomic series. Over this 27-month sample, the first principal component explains 52.1% of the total variance in the seven z-scored sub-indicators, indicating that more than half of the common variation is captured by a single factor.

Table 2 summarizes the PCA variance decomposition for the most recent quarterly recalibration.

Table 2: PCA variance explained (latest recalibration)

Component	Eigenvalue	Variance explained (%)
PC1	3.647	52.1
PC2	—	—
PC3–PC7	—	—

Note: Eigenvalues for PC2–PC7 omitted for brevity. The PC1 variance explained of 52.1% exceeds the 30% minimum target described in Section 8.

The backfilled series covers a period of notable sentiment variation, including post-reopening optimism in early 2023, property-sector stress in mid-2023, stimulus expectations in late 2024, and renewed trade-policy uncertainty in early 2025. These episodes provide useful variation for validating the index against external benchmarks.

5 Storage and Reproducibility

All data are stored in a SQLite database located at `data/index.db`. The database contains four tables:

1. `articles`: Raw scraped text with source, date, and URL metadata.
2. `article_sentiment`: Article-level sentiment scores from SnowNLP.
3. `monthly_index`: Monthly composite index values and all sub-indicator values.
4. `pca_params`: Archived PCA parameters (loadings, means, standard deviations, variance explained) for each recalibration date.

The full pipeline is deterministic given the same input data: fixing the scraped text, FRED series, and Google Trends data uniquely determines the index value.

6 Interpretation Guide

Table 3 provides a qualitative interpretation of the index.

Table 3: Index interpretation guide

Range	Interpretation
80–100	Strong optimism
60–80	Moderately positive
45–55	Neutral / mixed signals
20–45	Moderately negative
0–20	Crisis-level pessimism

7 Limitations

Several limitations should be noted when using the CESI.

1. **Sentiment model domain mismatch.** The SnowNLP library was trained on product reviews, not financial text. It may misclassify domain-specific expressions—for example, interpreting "interest rate cut" (降息) as negative due to the word "cut," when it is typically viewed positively in the current Chinese policy context.
2. **Manual keyword curation.** The positive, negative, and uncertainty keyword lists are manually curated and may not capture newly emerging topics or evolving policy terminology.
3. **Limited article depth.** News scrapers capture headlines and lead paragraphs only; full article text is not retrieved, potentially missing nuance.
4. **Google Trends geographic bias.** Google Trends data reflect search behavior on a platform with limited penetration in mainland China, and therefore primarily capture international attention.
5. **Linearity assumption.** The PCA approach assumes linear relationships among sub-indicators, which may not hold during structural breaks or regime changes.

8 Validation Approach

We assess index quality along four dimensions.

Internal consistency. We monitor the share of variance explained by PC1, targeting a minimum of 30%. We also compute Cronbach's alpha across the seven sub-indicators to assess internal reliability.

External benchmarking. The CESI is compared to established indicators including the Caixin/Markit Manufacturing PMI, the National Bureau of Statistics Consumer Confidence Index, and the Baker–Bloom–Davis China Economic Policy Uncertainty Index. We report contemporaneous correlations and lead–lag structures.

Predictive content. We conduct Granger causality tests to assess whether the CESI contains incremental information for forecasting future economic outcomes such as industrial production growth, retail sales growth, and equity market returns.

Stability. We perform leave-one-out analysis (dropping each sub-indicator in turn) and examine rolling-window loading stability to ensure that no single sub-indicator dominates the composite and that the factor structure is robust over time.